BLUESTOCK FINTECH

CAPSTONE PROJECT

MUTUAL FUND ANALYTICS PLATFORM

A Comprehensive Mutual Fund Analytics and Performance Evaluation System

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Prepared For

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Project Duration

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Submission Date

July 2026

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1. Introduction

The mutual fund industry has experienced rapid growth in recent years, attracting millions of investors and managing large volumes of financial data. With thousands of mutual fund schemes available across different categories, investors often find it challenging to evaluate and compare funds based on returns, risk, portfolio composition, and overall performance.

The Mutual Fund Analytics Platform was developed as part of the Bluestock Fintech Capstone Project to address these challenges. The project provides a comprehensive analytics solution that automates data ingestion, data cleaning, performance analysis, and interactive visualization using modern data analytics tools.

The platform leverages Python for data processing, SQLite for efficient data storage, SQL for querying, and Power BI for creating interactive dashboards. Advanced analytical techniques such as Sharpe Ratio, Value at Risk (VaR), Conditional Value at Risk (CVaR), portfolio concentration analysis using the Herfindahl-Hirschman Index (HHI), and a risk-based fund recommender system were implemented to provide meaningful insights for investors and analysts.

The primary objective of this project is to transform raw mutual fund data into actionable insights, enabling better investment decisions through data-driven analysis and visualization.

2. Problem Statement

Investors today have access to a wide variety of mutual fund schemes across different fund houses and investment categories. However, comparing these schemes based on returns, risk, expense ratio, portfolio allocation, and historical performance is often a complex and time-consuming process. The available information is usually spread across multiple sources, making it difficult for investors to make informed investment decisions.

To address this challenge, the **Mutual Fund Analytics Platform** was developed as an end-to-end analytics solution. The platform integrates multiple mutual fund datasets into a single system, performs automated data ingestion and cleaning, calculates key performance and risk metrics, and presents interactive dashboards for analysis. This enables investors and analysts to evaluate mutual funds more efficiently using reliable, data-driven insights.

The developed platform leverages Python, SQL, SQLite, and Power BI to transform raw mutual fund data into meaningful insights. Advanced analytics such as Sharpe Ratio, Value at Risk (VaR), Conditional Value at Risk (CVaR), portfolio concentration analysis using the Herfindahl-Hirschman Index (HHI), and a risk-based fund recommendation system help users evaluate fund performance, assess investment risk, and make informed financial decisions.

3. Objectives

- To develop an end-to-end Mutual Fund Analytics Platform for analyzing and evaluating mutual fund performance.

- To automate data ingestion, preprocessing, and cleaning using Python.

- To store processed data efficiently using SQLite for easy querying and analysis.

- To perform Exploratory Data Analysis (EDA) to identify trends, patterns, and insights from mutual fund data.

- To calculate important performance and risk metrics such as Sharpe Ratio, Value at Risk (VaR), and Conditional Value at Risk (CVaR).

- To analyze portfolio concentration using the Herfindahl-Hirschman Index (HHI).

- To develop a risk-based mutual fund recommendation system based on investor risk appetite.

- To build an interactive Power BI dashboard for effective visualization and decision-making.

- To generate actionable insights that help investors compare and evaluate mutual fund schemes.

4. Dataset Description

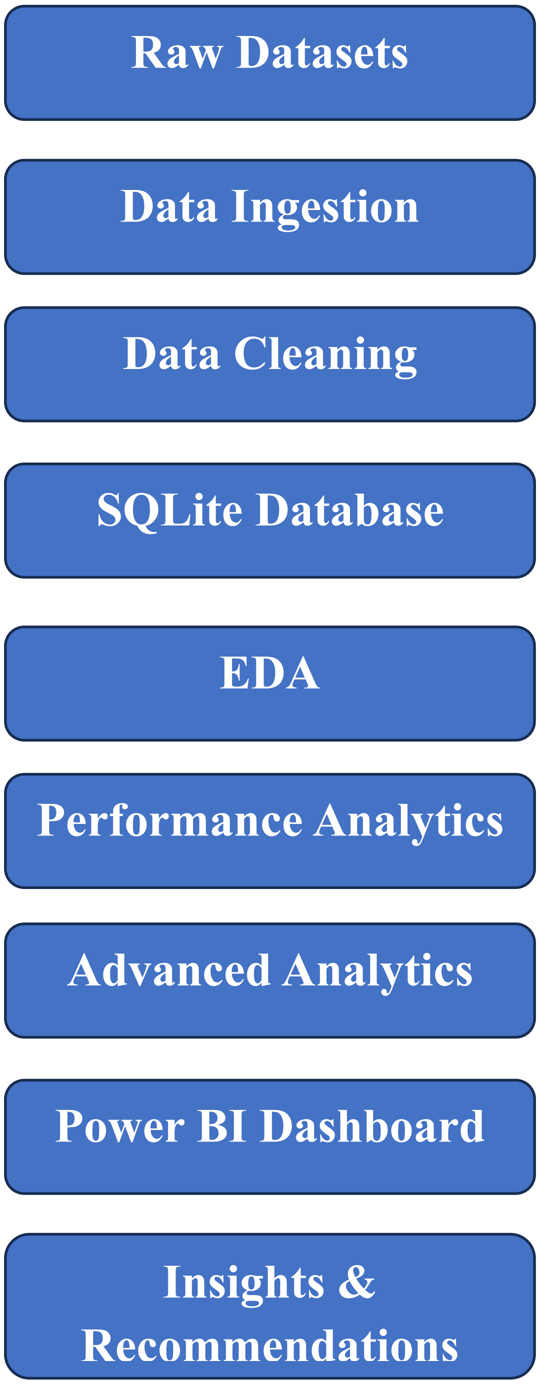
The Mutual Fund Analytics Platform utilizes multiple datasets collected from different sources to perform comprehensive analysis. These datasets contain information related to mutual fund schemes, NAV history, fund performance, investor transactions, portfolio holdings, and industry statistics. Each dataset serves a specific purpose in the analytics pipeline, enabling data cleaning, performance evaluation, risk analysis, and dashboard visualization.

Dataset Name	Description	Rows	Columns
01_fund_master.csv	Master information of all mutual fund schemes including fund house, category, benchmark, expense ratio, and fund manager details.	40	15
02_nav_history.csv	Historical daily Net Asset Value (NAV) data used for calculating daily returns, rolling performance, and risk metrics.	46,000	3
03_aum_by_fund_house.csv	Assets Under Management (AUM) details of different mutual fund houses over time.	90	5
04_monthly_sip_inflows.csv	Monthly Systematic Investment Plan (SIP) inflows, active SIP accounts, and SIP AUM statistics.	48	6
05_category_inflows.csv	Monthly net inflows and outflows across various mutual fund categories.	144	3

Dataset Name	Description	Rows	Columns
06_industry_folio_count.csv	Industry-wide mutual fund folio statistics across equity, debt, hybrid, and other investment categories.	21	6
07_scheme_performance.csv	Performance metrics of mutual fund schemes including returns, Sharpe Ratio, Beta, Alpha, Standard Deviation, and AUM.	40	19
08_investor_transactions.csv	Investor transaction records containing purchases, redemptions, SIPs, demographic information, and payment details.	32,778	13
09_portfolio_holdings.csv	Portfolio holdings of mutual fund schemes including stock names, sectors, portfolio weights, and market values.	322	8
10_benchmark_indices.csv	Historical benchmark index values used for comparing mutual fund performance with market indices	8,050	3

5. Project Architecture

The Mutual Fund Analytics Platform follows a structured end-to-end data analytics workflow. The project begins with collecting raw datasets, followed by data ingestion, preprocessing, data storage, analytics, and dashboard visualization. Each stage of the pipeline is designed to ensure data quality, efficient processing, and meaningful insights for decision-making.



6. Data Ingestion

The data ingestion process is the first stage of the Mutual Fund Analytics Platform. Multiple raw datasets in CSV format were imported into Python using the Pandas library. The datasets include mutual fund master data, NAV history, AUM details, SIP inflows, investor transactions, portfolio holdings, benchmark indices, and scheme performance data.

During the ingestion phase, each dataset was validated by examining its structure, including the number of rows, columns, data types, and sample records. Functions such as head(), shape(), info(), and describe() were used to verify data integrity before proceeding to the data cleaning stage.

Key Activities Performed :

- Imported all raw mutual fund datasets using the Pandas library.
- Verified dataset dimensions using shape().
- Examined column names and data types using info().
- Previewed sample records using head().
- Generated summary statistics using describe().
- Validated the datasets before data preprocessing.

7. Data Cleaning

Data cleaning was performed to improve data quality and ensure consistency across all datasets. Missing values, duplicate records, incorrect data types, and invalid values were identified and corrected before further analysis.

Date columns were converted into the appropriate datetime format, numerical fields were validated, and cleaned datasets were exported for analytical processing. This preprocessing step ensured accurate and reliable results throughout the project.

Cleaning Operations Performed :

- Checked for missing values and duplicate records.
- Converted date columns to the datetime format.
- Removed invalid records where applicable.
- Exported cleaned datasets for analytics and dashboard development.

8. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the mutual fund datasets and identify meaningful trends, distributions, and relationships. Various statistical summaries and visualizations were generated to analyze fund categories, risk levels, expense ratios, Assets Under Management (AUM), and historical performance. The insights obtained during this phase served as the foundation for performance evaluation and advanced analytics.

8.1 Daily NAV Trend Analysis

The daily Net Asset Value (NAV) trends of mutual fund schemes were analyzed to understand their historical performance over time. The visualization highlights the growth patterns and fluctuations in NAV values across different schemes, providing insights into long-term fund performance and market behaviour.

Observation

The analysis indicates that most mutual fund schemes exhibited a positive long-term growth trend, although the rate of growth varied across schemes. Certain funds demonstrated higher volatility, reflecting the impact of changing market conditions and investment strategies.

8.2 Fund House Analysis

The mutual fund schemes were analyzed based on their respective fund houses to understand the distribution of schemes across different Asset Management Companies (AMCs). This analysis provides insights into the diversity of fund offerings and the contribution of each fund house within the dataset.

Observation

The dataset contains schemes from multiple leading fund houses, ensuring a balanced representation of the Indian mutual fund industry for analysis.

8.3 Risk Category Analysis

Risk categories assigned to mutual fund schemes were examined to understand the overall distribution of investment risk. The analysis helps investors identify schemes that align with their financial goals and risk tolerance.

Observation

The majority of schemes belong to Moderate and High-risk categories, while relatively fewer schemes fall under the Low-risk category.

9.Performance Analytics

Performance Analytics was conducted to evaluate the historical returns and risk-adjusted performance of mutual fund schemes. Various financial metrics were analyzed to compare the performance of different funds and identify investment opportunities based on return, volatility, and risk.

9.1.Return Analysis

The returns generated by mutual fund schemes over different investment periods were analyzed to evaluate their overall performance. Comparing returns across schemes helps investors identify funds that have consistently delivered better growth.

Observation

Several equity-oriented mutual fund schemes generated higher long-term returns compared to debt-oriented schemes**.**

9.2 Risk-Adjusted Performance

Risk-adjusted performance metrics were analyzed to determine whether the returns generated by mutual funds were proportional to the level of risk undertaken by investors.

Observation

Funds with higher risk-adjusted performance demonstrated a better balance between returns and investment risk**.**

9.3 Volatility Analysis

The volatility of mutual fund schemes was evaluated to understand the stability of returns over time. This analysis helps investors assess the consistency of fund performance.

Observation

Equity-focused schemes exhibited comparatively higher volatility, whereas debt-oriented schemes showed more stable performance.

10.Advanced Analytics

Advanced Analytics was performed to gain deeper insights into mutual fund performance using quantitative financial metrics. These techniques help evaluate risk-adjusted returns, portfolio concentration, downside risk, and investor suitability for different mutual fund schemes.

10.1 Sharpe Ratio Analysis

The Sharpe Ratio was used to measure the risk-adjusted performance of mutual fund schemes by comparing excess returns with total investment risk. A higher Sharpe Ratio indicates better returns for each unit of risk undertaken.

Observation

Funds with higher Sharpe Ratios demonstrated superior risk-adjusted performance and were considered more efficient investment options.

10.2 Value at Risk (VaR)

Value at Risk (VaR) was calculated to estimate the maximum expected loss over a specified confidence level under normal market conditions.

Observation

VaR analysis highlighted the potential downside risk associated with each mutual fund scheme.

10.3 Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR) was used to estimate the average loss beyond the Value at Risk threshold, providing a more comprehensive measure of downside risk.

Observation

CVaR offered a better understanding of extreme market losses compared to VaR alone.

10. Mutual Fund Recommendation System

A rule-based recommendation system was developed to suggest suitable mutual fund schemes based on the investor's risk appetite and fund characteristics.

Observation

The recommendation system assists investors in identifying funds that align with their investment objectives and risk tolerance.

11. Power BI Dashboard

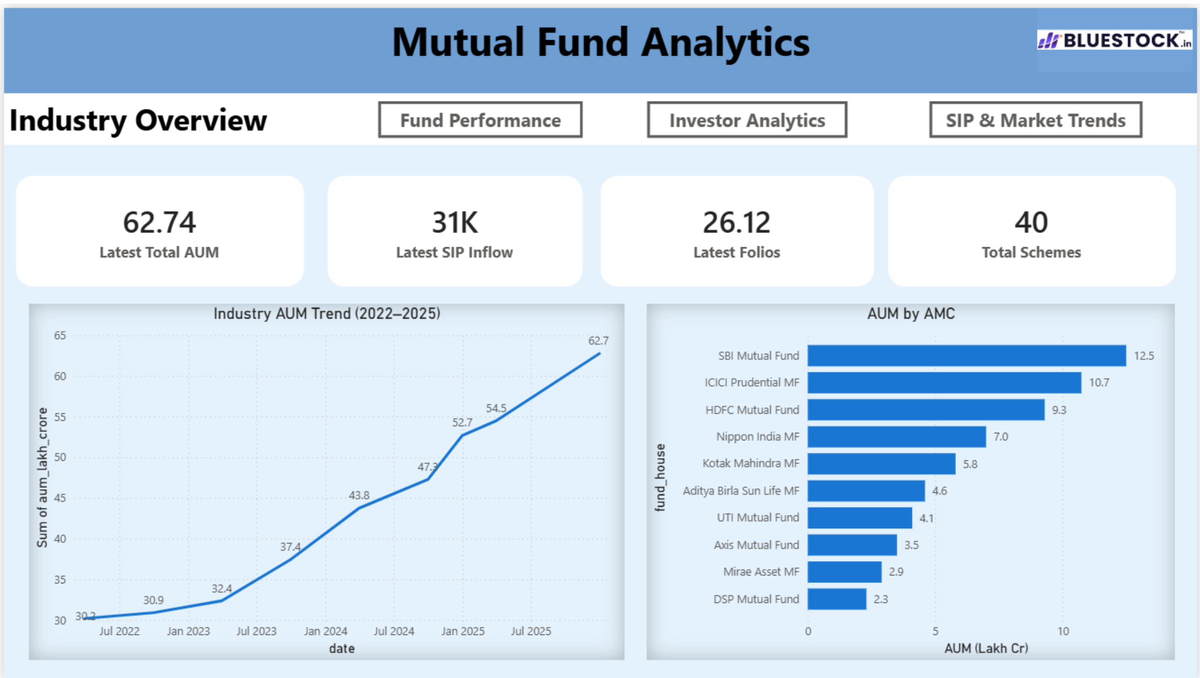
An interactive Power BI dashboard was developed to visualize mutual fund performance, investment trends, portfolio statistics, and key financial metrics. The dashboard enables users to explore the data efficiently and supports better investment decision-making through interactive visualizations.

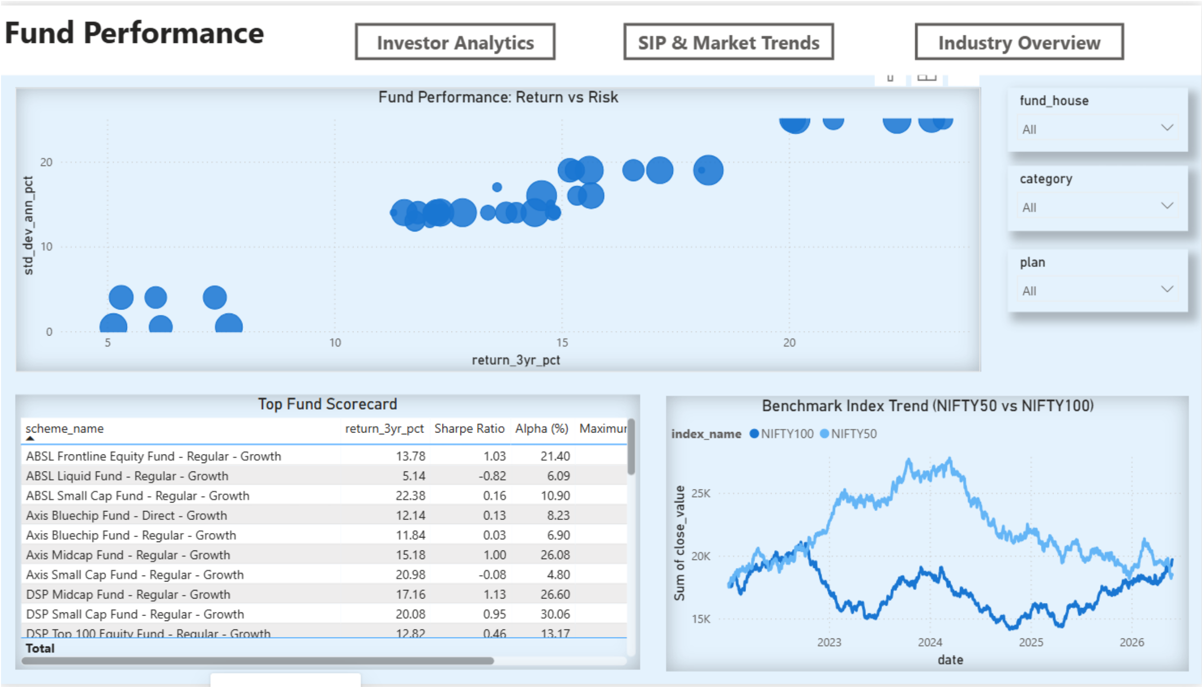
Dashboard Features

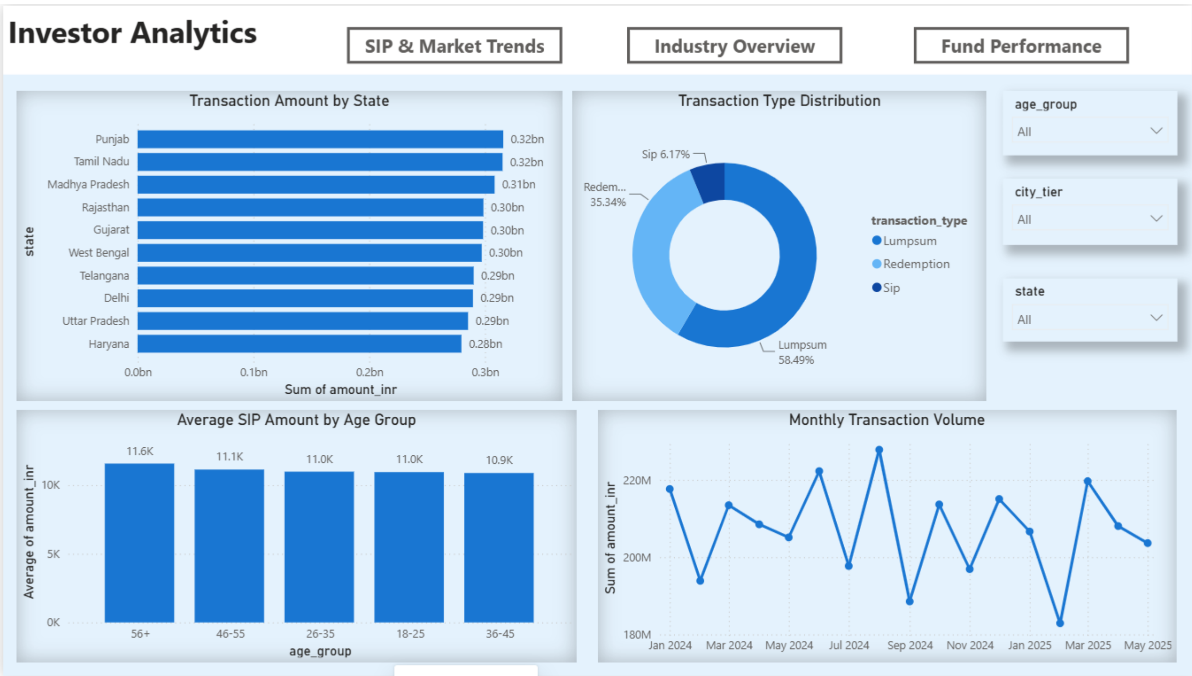
- KPI Cards displaying Total AUM, Total Schemes, Average Return, and Average Expense Ratio.
- Fund-wise and Category-wise performance comparison.
- Risk category and expense ratio analysis.
- Interactive filters for Fund House, Category, and Scheme.
- Time-series visualizations for historical NAV trends.

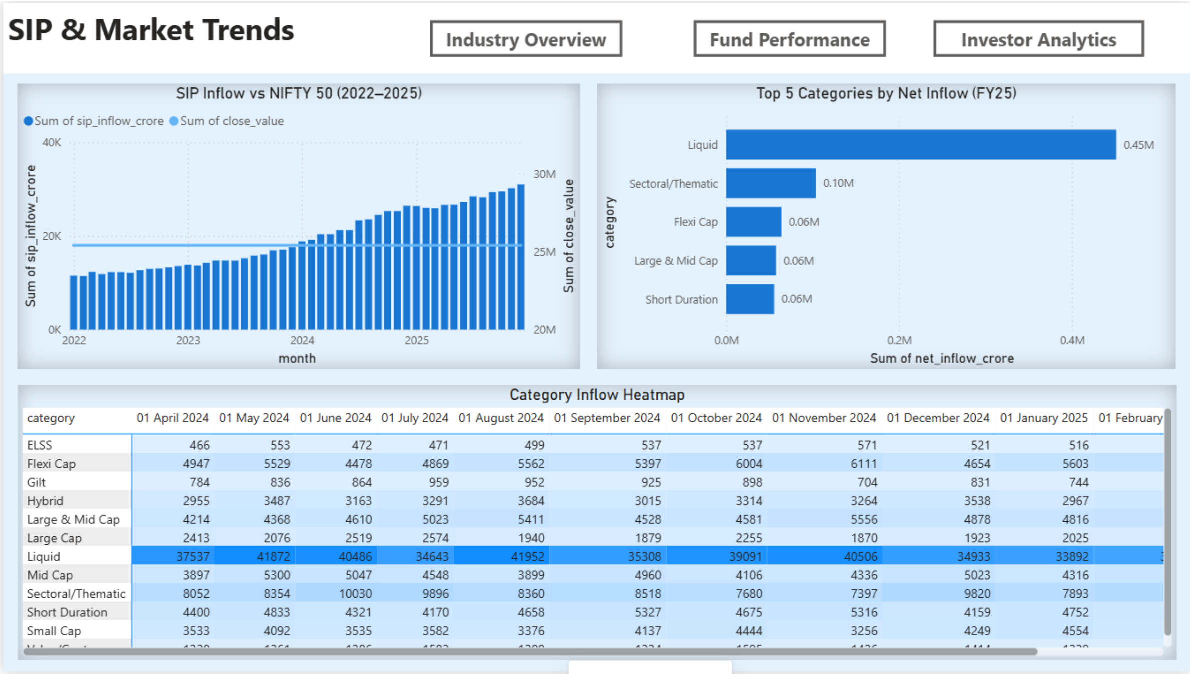
Key Insights

- Equity-oriented funds generated comparatively higher long-term returns.
- Risk-adjusted metrics helped identify consistently performing schemes.
- Expense ratios varied across different categories of mutual funds.
- Dashboard filters enabled dynamic comparison between multiple schemes.
- Interactive visualizations simplified investment analysis.




Fund Performance
Investor Analytics
SIP & Market Trends
Industry Overview
Fund Performance: Return vs Risk
std_dev_ann_pct
return_3yr_pct
fund_house
All
category
All
plan
All
Top Fund Scorecard
scheme_name return_3yr_pct Sharpe Ratio Alpha (%) Maximur
ABSL Frontline Equity Fund - Regular - Growth 13.78 1.03 21.40
ABSL Liquid Fund - Regular - Growth 5.14 -0.82 6.09
ABSL Small Cap Fund - Regular - Growth 22.38 0.16 10.90
Axis Bluechip Fund - Direct - Growth 12.14 0.13 8.23
Axis Bluechip Fund - Regular - Growth 11.84 0.03 6.90
Axis Midcap Fund - Regular - Growth 15.18 1.00 26.08
Axis Small Cap Fund - Regular - Growth 20.98 -0.08 4.80
DSP Midcap Fund - Regular - Growth 17.16 1.13 26.60
DSP Small Cap Fund - Regular - Growth 20.08 0.95 30.06
DSP Top 100 Equity Fund - Regular - Growth 12.82 0.46 13.17
Total
Benchmark Index Trend (NIFTY50 vs NIFTY100)
index_name NIFTY100 NIFTY50
Sum of close_value
date


Investor Analytics
SIP & Market Trends
Industry Overview
Fund Performance
Transaction Amount by State
Punjab 0.32bn
Tamil Nadu 0.32bn
Madhya Pradesh 0.31bn
Rajasthan 0.30bn
Gujarat 0.30bn
West Bengal 0.30bn
Telangana 0.29bn
Delhi 0.29bn
Uttar Pradesh 0.29bn
Haryana 0.28bn
state
Sum of amount_inr
Transaction Type Distribution
Sip 6.17%
Redem... 35.34%
Lumpsum 58.49%
transaction_type
Lumpsum
Redemption
Sip
age_group
All
city_tier
All
state
All
Average SIP Amount by Age Group
11.6K 11.1K 11.0K 11.0K 10.9K
56+ 46-55 26-35 18-25 36-45
Average of amount_inr
age_group
Monthly Transaction Volume
Sum of amount_inr
Jan 2024 Mar 2024 May 2024 Jul 2024 Sep 2024 Nov 2024 Jan 2025 Mar 2025 May 2025



12. Technologies Used

Technology	Purpose
Python	Data Processing and Analysis
Pandas	Data Cleaning and Manipulation
NumPy	Numerical Computations
SQLite	Database Management
SQL	Data Querying
Power BI	Dashboard Development
Jupyter Notebook	Data Analysis Environment
Git & GitHub	Version Control and Project Repository

13. Conclusion

The Mutual Fund Analytics Platform successfully integrates data engineering, financial analytics, and business intelligence into a single solution. The project demonstrates how raw mutual fund data can be transformed into meaningful insights through data cleaning, exploratory analysis, performance evaluation, and advanced analytics.

The developed platform enables investors to compare mutual fund schemes, evaluate investment risk, and make informed financial decisions. The interactive Power BI dashboard further enhances usability by presenting complex financial information through intuitive visualizations.

14. Future Scope

- Integrate live NAV data using APIs.
- Develop machine learning models for mutual fund return prediction.
- Enhance the recommendation system using AI techniques.
- Deploy the project as a web application.
- Add portfolio optimization and investment simulation features.

15. References

- AMFI India (Association of Mutual Funds in India)
- NSE India
- BSE India
- Python Documentation
- Pandas Documentation
- SQLite Documentation
- Microsoft Power BI Documentation